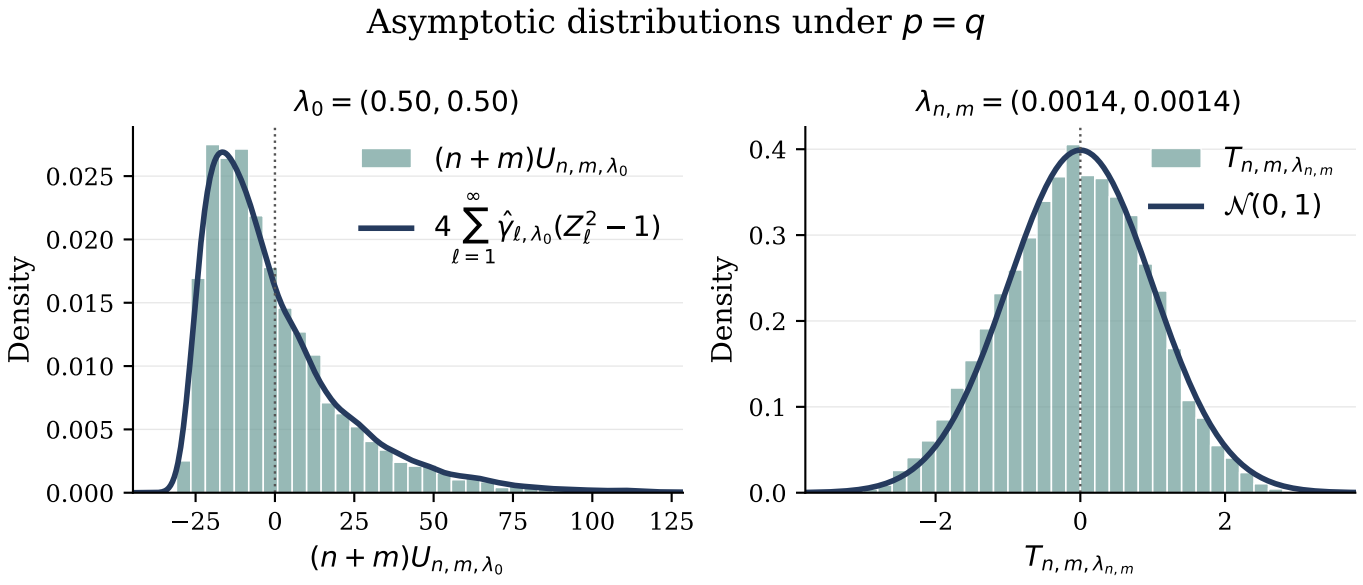
Asymptotic distributions under $p = q$
$\lambda_0 = (0.50, 0.50)$
$(n+m)U_{n,m,\lambda_0}$
$4\sum_{\ell=1}^{\infty}\hat{\gamma}_{\ell,\lambda_0}(Z_\ell^2 - 1)$
Density
0.025
0.020
0.015
0.010
0.005
0.000
−25
0
25
50
75
100
125
$(n+m)U_{n,m,\lambda_0}$
$\lambda_{n,m} = (0.0014, 0.0014)$
$T_{n,m,\lambda_{n,m}}$
$\mathcal{N}(0,1)$
Density
0.4
0.3
0.2
0.1
0.0
−2
0
2
$T_{n,m,\lambda_{n,m}}$